

ConfDiff: Confidence-Guided Representation Diffusion for Video Moment Retrieval

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Abstract

Video moment retrieval (VMR) aims to localize query-relevant temporal segments in untrimmed videos and estimate reliable relevance scores. While recent transformer-based frameworks have achieved promising performance, most still adopt deterministic boundary prediction, making them insufficient for modeling boundary ambiguity and prone to positional bias in temporal representations. We present ConfDiff, a confidence-guided diffusion framework that treats segment-level temporal representations as stochastic variables and refines them through progressive denoising. Our method first aligns video and text features to build a confidence correlation map, which combines frame-level saliency and cross-modal similarity into a soft temporal prior. This prior guides selective noise injection into temporally ambiguous, low-confidence regions, enabling the denoiser to focus refinement on uncertain intervals while preserving reliable evidence. The resulting uncertainty-aware representations are then decoded by a query-guided head to produce precise temporal boundaries and calibrated relevance scores, jointly benefiting moment retrieval and highlight detection (HD). In addition, we introduce an output-space consistency objective between denoised and clean predictions to better couple generative refinement with discriminative decoding, leading to more stable training and improved generalization. Experiments on QVHighlights, Charades-STA, and TACoS demonstrate consistent gains over strong baselines across both VMR and HD tasks.

1. Introduction

Video moment retrieval (VMR) aims to localize, within an untrimmed video, the temporal segment that semantically corresponds to a natural-language query. With the rapid growth of online video and the increasing demand for multi-

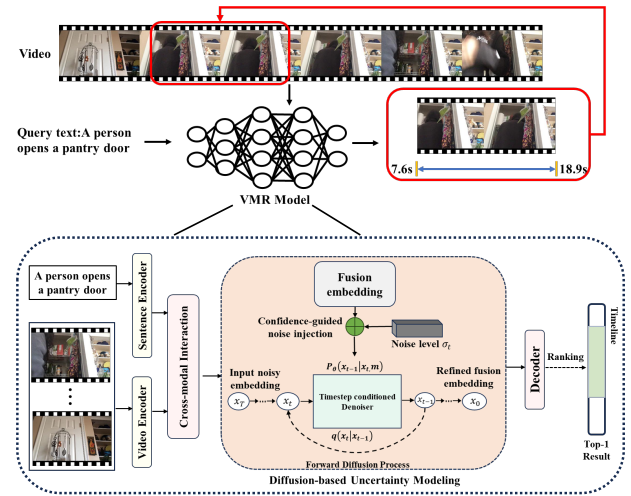


Figure 1. Overview of ConfDiff for video moment retrieval. Given an untrimmed video and a natural-language query, ConfDiff encodes both modalities to form a fusion embedding over the timeline, then uses confidence-guided diffusion to refine ambiguous regions before a decoder ranks temporal locations and outputs the top-1 moment.

modal retrieval, VMR has become a key primitive for video understanding, content-based search, and human-computer interaction, and is gradually evolving into a core building block for long-form video analysis [24, 47]. Despite substantial progress from regression- and proposal-based approaches, a central bottleneck remains unresolved: temporal event boundaries in the wild are inherently ambiguous and uncertain [10, 41]. Unlike the relatively crisp geometric contours of objects in images, event onsets and offsets in video are often gradual and context-dependent, and are further influenced by narrative pacing, shot transitions, action latency, and the generality or ambiguity of language descriptions. As a result, human annotations are inevitably subjective and may reflect multiple plausible yet incompatible boundary choices.

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Recent efforts attempt to mitigate boundary ambiguity via contrastive learning [45, 50], cross-modal attention [8], Bayesian modeling [11], or ensembling [9]. However, most existing methods [2, 25, 33, 40] still operate within a deterministic point-prediction paradigm: they either directly regress start/end coordinates or select a single boundary pair from a candidate set. Such designs implicitly assume that there exists one “correct” boundary pair, and thus fail to reflect the continuous and potentially multi-modal set of feasible boundaries over time. We argue that boundary ambiguity is, in essence, a probability distribution over the timeline rather than an isolated pair of points. Under this view, point-to-point regression struggles to capture the context-dependent distributional structure underlying true boundaries, and becomes particularly fragile in scenarios with smooth semantic transitions, weak visual cues, or cross-shot continuity—often leading to boundary drift, miscalibrated confidence, and reduced robustness under distribution shift.

We revisit VMR from a generative perspective and reformulate the temporal localization problem as a diffusion-style denoising process in the space of temporal representations (Fig. 1). Instead of directly regressing fixed boundary points, the model performs multi-step denoising on a noisy and uncertain latent representation of the target event, gradually converging to a distribution-aware representation that explicitly encodes boundary uncertainty. Specifically, we propose ConfDiff, a diffusion-based framework that (i) encodes text queries and video frames into aligned cross-modal embeddings; (ii) builds a cross-attentive Transformer to obtain query-aware video memory representations and a multi-layer cross-modal correlation graph, from which it derives a frame-level confidence curve that characterizes alignment strength along the timeline, localizes low-confidence regions, and serves as an uncertainty-aware budget for generative refinement; and (iii) explicitly constrains the diffusion process to low-confidence temporal regions, injecting noise according to a cosine schedule and conditioning the denoising network on the timestep, thereby progressively “recovering” clean and distribution-aware representations. To tightly couple the generative branch with the discriminative decoder, we share a single decoder between the “clean” and “reconstructed” memories and impose an output-space consistency loss only on matched positive queries. This unified design stabilizes optimization, keeps the diffusion dynamics aligned with the VMR objective, and reduces the distribution gap between generative reconstruction and discriminative decoding.

In summary, we (i) cast VMR temporal refinement as generative diffusion over the representation space, explicitly treating boundary ambiguity as a time-varying distribution rather than a deterministic boundary pair; (ii) develop ConfDiff, which derives a frame-level confidence map from cross-modal attention and performs confidence-gated, se-

lective diffusion with small-step deterministic refinement at inference; and (iii) couple the generative reconstruction branch with a query-based discriminative decoder discriminative decoder via a lightweight output-space consistency loss, leading to state-of-the-art performance on QVHighlights, Charades-STA, and TACoS, particularly under fuzzy and weakly defined event boundaries.

2. Related Work

2.1. Video Moment Retrieval

VMR aims to localize, within an untrimmed video, the temporal segment that matches a natural-language query. Existing methods can be roughly divided into two categories. (1) Pre-defined video proposal paradigms [6, 22, 44, 44, 48] typically first densely sample candidate segments along the temporal axis using sliding windows or dedicated proposal generation networks, and then rank these candidates by their cross-modal similarity to the query in order to select high-scoring moments. (2) Learnable proposal/query methods [12, 18–20, 24, 32, 39] introduce learnable queries or proposal heads that adapt to data statistics and perform one- or two-stage refinement. Representative “classic” works include Moment-DETR [16], which formulates VMR as set prediction with end-to-end coordinate and saliency regression, and CG-DETR [31], which injects relevance guidance into cross-modal attention to suppress distractors. Building on this line, UVCOM [42] jointly models retrieval and highlight detection, MS-DETR [29] decouples motion and semantics to co-optimize the two tasks, and BM-DETR [14] leverages background awareness and negative queries to alleviate weak alignment, thereby further unifying tasks and strengthening boundary modeling. However, learnable proposal/query models still explicitly fit positional distributions and are prone to absorbing dataset-specific temporal priors, which limits cross-domain transfer. In contrast, our approach avoids handcrafted positional priors via position-agnostic initialization and conditional modeling; under cross-modal alignment and consistency regularization, the model progressively converges to the target segment, thereby yielding more interpretable uncertainty.

2.2. Diffusion models

Diffusion models provide a controllable generative trajectory from noise to data: training corrupts data via a forward process while learning the reverse dynamics; inference then starts from a prior and iteratively denoises, naturally supporting multi-modality and principled uncertainty. This paradigm has been validated in structured prediction for detection/localization: DiffusionDet [3] reframes object detection as multi-step refinement from noisy boxes to target boxes, enabling variable-cardinality prediction with progressive quality gains; DiffusionInst [7] replaces an-

chors/proposals with “noise $\rightarrow$ instance-aware filters,” reducing hand-crafted priors and supporting single- or multi-step inference. In video understanding and temporal modeling, the same idea applies: DiffusionTrack [27] casts single-object tracking as point-set denoising for more robust cross-frame localization, and DiffMOT [28] leverages a decoupled, diffusion-based motion predictor to capture nonlinear trajectories while balancing accuracy and runtime. In more temporally demanding action/event scenarios, DiffTAD [34] treats temporal proposals as stochastic intervals and learns their reverse denoising, showing that iterative refinement stabilizes boundaries on long-video benchmarks. Taken together, the diffusion paradigm offers a calibratable coarse-to-fine generator and a general route for modeling boundary uncertainty over long sequences and across modalities.

2.3. A Diffusion Perspective on Video Moment Retrieval

In recent years, diffusion paradigms have been introduced to VMR, using a “from noise to segment” iterative generation process to explicitly model temporal uncertainty and the multi-modality of plausible segments. MomentDiff [17] starts from stochastic intervals and progressively denoises toward the semantic target, alleviating positional bias and dependence on proposals; DiffusionVMR [49] formulates moment retrieval and highlight detection within a unified, conditionally denoised framework; for cross-domain and unseen semantics, FVE [26] employs text-guided video diffusion to synthesize “pseudo segments” in the target domain, improving generalization. In the closely related task of activity localization, DenoiseLoc [43] injects noise into candidate boundaries and learns to denoise them, demonstrating the robustness of “boundary denoising”; additionally, CVDN [21] adopts sentence-conditioned denoising for fine-grained cross-modal alignment. However, most existing efforts denoise in the coordinate/proposal space, leaving the uncertainty in the representation space underexplored. Motivated by this gap, we avoid any fixed proposal prior and instead drive boundary estimates to converge controllably via representation-space conditioning, yielding calibrated, uncertainty-aware temporal representations.

3. Method

3.1. Overview

Given a video $V = [v_1, \dots, v_{L_v}]$ discretized into L_v segments and a natural-language query $Q = [q_1, \dots, q_{L_q}]$ with L_q tokens, VMR aims to learn a model Ω that predicts (i) a per-segment relevance sequence $\mathbf{s} = [s_1, \dots, s_{L_v}]$ and (ii) the target moment parameters (m_c, m_σ) . Here, s_i encodes the matching strength between segment v_i and the query, m_c approximates the temporal center of the moment, and

m_σ measures its temporal extent/scale (i.e., the span of the moment) [31]. The overall architecture is shown in Fig. 2.

The core objective of ConfDiff is to model temporal boundary uncertainty in a generative manner. To this end, Section 3.2 constructs a cross-attentive correlation graph between video and text embeddings, yielding a frame-level confidence map that quantifies cross-modal alignment strength and highlights ambiguous regions. Section 3.3 introduces a diffusion-based refinement process operating in representation space, where noise is selectively injected into low-confidence regions and progressively denoised into distribution-aware event representations that capture boundary ambiguity and multi-modal semantics. Section 3.4 further incorporates a consistency-alignment objective that couples generative refinement with discriminative decoding.

3.2. Cross-Attentive Transformer Encoder

To capture both global semantics and temporal motion, we freeze a pre-trained CLIP [36] visual encoder to extract appearance descriptors and employ SlowFast [5] to capture spatiotemporal cues. The two streams are concatenated along the clip dimension, yielding the clip-level sequence $\mathbf{E}_V = [\mathbf{e}_0, \dots, \mathbf{e}_{L_v-1}] \in \mathbb{R}^{L_v \times D_v}$. On the text side, CLIP-Text produces token-level embeddings $\mathbf{E}_Q \in \mathbb{R}^{L_q \times D_q}$. Two independent MLPs project video and text features into a shared space of dimension D , giving $\mathbf{V} = \text{MLP}_v(\mathbf{E}_V) \in \mathbb{R}^{L_v \times D}$ and $\mathbf{T} = \text{MLP}_q(\mathbf{E}_Q) \in \mathbb{R}^{L_q \times D}$. To enhance cross-modal interaction and fully aggregate sentence cues, we adopt a cross-attention transformer that takes video tokens as queries and text tokens as keys/values. The query-based video representations can be computed by:

$$\tilde{\mathbf{V}} = \text{MHA}(\underbrace{\mathbf{V}\mathbf{W}_Q}_{\mathbf{Q}_v}, \underbrace{\mathbf{T}\mathbf{W}_K}_{\mathbf{K}_t}, \underbrace{\mathbf{T}\mathbf{W}_V}_{\mathbf{V}_t}). \quad (1)$$

$\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ are learnable projection matrices mapping $\mathbf{V}$ and $\mathbf{T}$ to the query/key/value spaces respectively.

3.3. Frame-Level Confidence Map Construction

Video moment boundaries typically arise in semantic transition zones (e.g., gradual action onsets/offsets, narrative shot junctions, and cross-frame semantic transfer). In these regions, cross-modal evidence exhibits low consistency and high uncertainty. Applying generative refinement uniformly across the entire timeline introduces unnecessary computation and can over-correct high-confidence areas. Therefore, for a video discretized into L_v frames, we construct a frame-level confidence curve $\mathbf{c} = [c_1, \dots, c_{L_v}]$ from cross-modal attention. This curve maps the text–video alignment strength to a comparable, thresholdable temporal function, which we use to localize uncertain regions and to gate subsequent noise-injection–denoising and multi-step refinement.

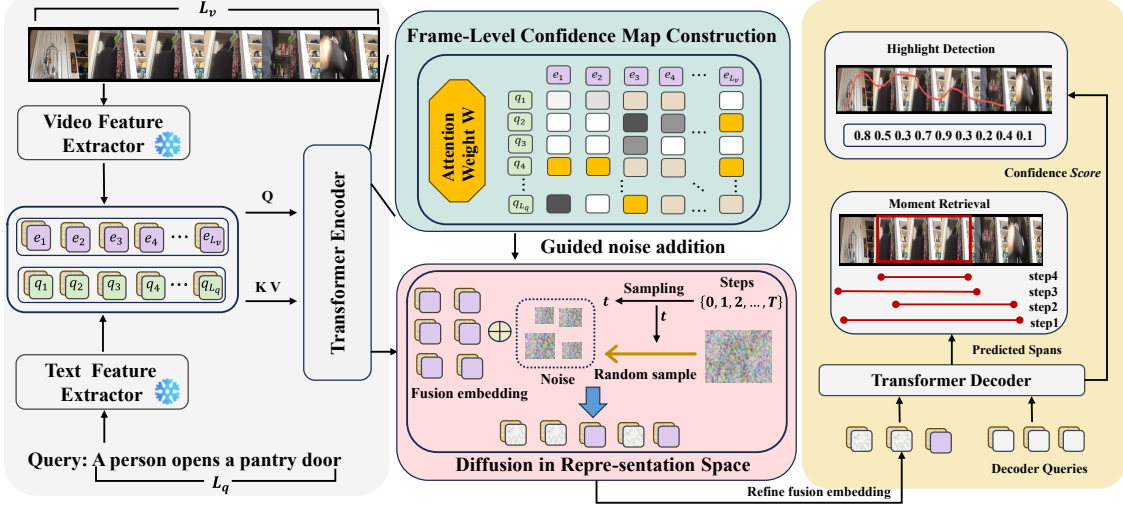


Figure 2. Overview of ConfDiff. Given a video and a natural-language query, frozen video and text encoders first extract clip- and token-level features, which are fused by a cross-attentive transformer encoder into a shared memory (left). Aggregated cross-modal attention yields a frame-level confidence map highlighting temporally ambiguous, low-confidence regions (top-middle). A diffusion module then injects and removes noise only in these regions, focusing refinement on uncertain boundaries and producing distribution-aware event features (bottom-middle). Finally, a transformer decoder consumes the refined memory to jointly perform highlight detection and moment retrieval, outputting frame-wise confidence scores and temporal spans (right).

Frame-Level Confidence Estimation from Cross-Modal Attention. Text queries typically exhibit a multi-level structure (e.g., subject, predicate, object), and their cross-modal alignment evidence is often dispersed across query tokens. To consolidate these signals into a thresholdable frame-level statistic, we sum along the token dimension to obtain a per-frame semantic support vector.

$$\mathbf{a}^{(l,h)} = \sum_{j=1}^{L_q} \mathbf{A}_{:,j}^{(l,h)}, \quad (2)$$

Among them, $\mathbf{A}^{(l,h)} \in \mathbb{R}^{L_v \times L_q}$ denotes the cross-modal attention matrix, representing the alignment between text tokens and video frames at the l -th layer and the h -th head. By summing over all tokens, we derive a frame-level semantic support vector $\mathbf{a}^{(l,h)}$, which aggregates the alignment information of each frame with all text tokens. This aggregation mitigates the reliance on individual tokens and yields a more stable frame-level confidence curve. Since each attention head may introduce noise or bias, we further integrate evidence across all layers and heads to reduce variance and stabilize the confidence estimation.

$$\mathbf{c}_{\text{raw}} = \frac{1}{LH} \sum_{l=1}^L \sum_{h=1}^H \mathbf{a}^{(l,h)} \in \mathbb{R}^{L_v}, \quad (3)$$

Here, L denotes the number of cross-modal layers, and H represents the number of attention heads. This averaging process fuses multi-scale information from shallow to

deep layers as well as multi-view cues from different heads, thereby reducing variance and suppressing abnormal peaks. However, the absolute magnitude of c_{raw} is affected by the video length L_v and the overall attention quality, making it incomparable across samples. To address this, we subsequently introduce a length-aware normalization.

$$\tilde{c}_i = \frac{L_v c_i^{\text{raw}}}{\sum_{j=1}^{L_v} c_j^{\text{raw}} + \epsilon L_v}, \quad i = 1, \dots, L_v, \quad (4)$$

Here, $\epsilon > 0$ is a numerical stability parameter. After this normalization, we obtain a length-independent confidence curve $\tilde{c}$. To enable the setting of a unified threshold and ensure more stable cross-video comparisons, we apply max-normalization to the normalized confidence curve, mapping it to the range $[0, 1]$, resulting in the relative confidence $\hat{c} \in [0, 1]^{L_v}$. Next, we use the median of each video as the threshold to distinguish high-confidence from low-confidence frames. The median is less sensitive to outliers and provides a robust way to categorize video frames into low-confidence and high-confidence classes.

$$b = \text{median} \{ \hat{c}_i \}_{i=1}^{L_v}. \quad (5)$$

Based on this criterion, we define the set of low-confidence frames as $\mathcal{I}_{\text{low}}$, denoted as:

$$\mathcal{I}_{\text{low}} = \{ i \in \{1, \dots, L_v\} \mid \hat{c}_i < b \}, \quad (6)$$

These frames correspond to regions with high uncertainty. Subsequent generative refinement (including noise injection

and denoising, as well as multi-step fine-tuning) will be specifically applied to these regions, without over-adjusting high-confidence areas.

Budgeted Stochastic Gating. During training, we apply budgeted stochastic gating over $\mathcal{I}_{\text{low}}$ to control computational cost and introduce stochastic regularization:

$$\begin{aligned} m_i &\sim \text{Bernoulli}(p), \\ \mathcal{M} &= \{i \in \mathcal{I}_{\text{low}} \mid m_i = 1\}, \\ \mathbb{E}[|\mathcal{M}|] &= p|\mathcal{I}_{\text{low}}|. \end{aligned} \quad (7)$$

At each iteration, we independently subsample the low-confidence set $\mathcal{I}_{\text{low}}$ by drawing each index with probability p , thereby forming a stochastic mask $\mathcal{M}$ on which generative repair is applied. This selective diffusion concentrates computation on uncertain regions, implicitly regularizes training by randomizing the updated locations over time, and tightly couples noise injection and denoising with the model’s own confidence estimates.

During inference, we dispense with random sampling and instead apply small multi-step updates deterministically within the low-confidence region. The step size is adaptively scaled by $1 - \hat{c}_i$, ensuring that the model refines areas where the certainty is lower, in accordance with the principle ”refine more where less certain.”

3.4. Low-Confidence-Guided Diffusion in Representation Space

To avoid unnecessary corrections on high-confidence frames, we perform selective diffusion in the representation space based on Sec. 3.3: noise injection, denoising, and refinement are applied only to low-confidence positions. This strategy concentrates generative freedom on ambiguous regions while preserving the stability of confident frames, leading to more reliable boundary fitting and uncertainty modeling.

Scheduling and Timestep Embedding. Let T be the number of diffusion steps. We adopt the discrete cosine schedule (Nichol & Dhariwal) to obtain $\{\beta_t\}_{t=1}^T$ and define

$$\bar{\alpha}_t = \prod_{k=1}^t (1 - \beta_k), \quad \sqrt{\bar{\alpha}_t}, \sqrt{1 - \bar{\alpha}_t} \in (0, 1]. \quad (8)$$

Let the ”clean” memory representation be $\mathbf{x}_0 \in \mathbb{R}^{L_v \times D}$ (the query-based video representations introduced in Sec. 3.2). The forward noising at step t is

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (9)$$

The timestep is encoded by a sinusoidal embedding $\gamma(t) \in \mathbb{R}^D$ that conditions the denoising head via FiLM [35] (feature-wise scale/shift) on LayerNorm’ed features, jointly with the cross-modal condition so that denoising remains text- and context-aware.

Training: Masked Noising and Noise Regression. During training we inject noise only at low-confidence positions. Let $\mathcal{M} \subseteq \mathcal{I}_{\text{low}}$ be the budgeted subset sampled as in Sec. 3.3. For $i \in \mathcal{M}$ we form $\mathbf{x}_t^{(i)}$ by Eq. (8); for $i \notin \mathcal{M}$ we keep $\mathbf{x}_0^{(i)}$ unchanged. With denoiser $\epsilon_\theta(\cdot)$, we minimize a masked noise-prediction loss:

$$\mathcal{L}_{\text{diff}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\epsilon_\theta(\mathbf{x}_t^{(i)}, \gamma(t), \tilde{\mathbf{V}}) - \boldsymbol{\epsilon}^{(i)}\|_2^2. \quad (10)$$

This objective enables the model to remove injected noise only in regions with uncertain alignment, while $\tilde{\mathbf{V}}$ ensures that the denoising process remains consistent with cross-modal semantics.

To reduce the gap between the discriminative and generative distributions, we further construct a standard reconstruction process.

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, \gamma(t), \tilde{\mathbf{V}})}{\sqrt{\bar{\alpha}_t}}, \quad (11)$$

Rather than simply restoring the original feature distribution, this reconstruction acts as a task-oriented refinement: it steers the representation toward a feature manifold that is more favorable for the VMR task, while still remaining consistent with the underlying diffusion dynamics. With a probability, $\hat{\mathbf{x}}_0$ is further fed into the decoder to obtain an auxiliary ”refined” output. This auxiliary branch, used in Sec. 3.5 for consistency alignment, provides a mild regularization toward the VMR-specific representation, whereas the main output remains derived from the unrefined memory.

Inference : Uncertainty-Guided Small-Step Refinement. During the inference stage, instead of performing full reverse diffusion, we adopt a lightweight deterministic refinement process over a short noise range (e.g., $t \in \{10 \rightarrow 2\}$) for N_{steps} iterations. In each iteration, we first synthesize a noised version $\mathbf{x}_t$ from the current representation $\mathbf{x}_0$, optionally omitting the noise term for stability. The corresponding denoised estimate $\hat{\mathbf{x}}_0$ is then computed using Eq. (10).

To prevent overcorrection, updates are restricted to low-confidence regions $\mathcal{I}_{\text{low}}$ based on an uncertainty-aware interpolation strategy:

$$\begin{aligned} \mathbf{x}_0^{(i)} &\leftarrow (1 - u_i) \mathbf{x}_0^{(i)} + u_i \hat{\mathbf{x}}_0^{(i)}, \\ u_i &= \frac{u_{\text{base}}}{N_{\text{steps}}} (1 - \hat{c}_i) \mathbf{1}[i \in \mathcal{I}_{\text{low}}], \end{aligned} \quad (12)$$

where $\hat{c}_i$ denotes the normalized confidence score, $u_{\text{base}} \in (0, 1)$ controls the global update strength, and $\mathbf{1}[\cdot]$ is the indicator function. This strategy ensures selectivity (only uncertain positions are updated) and budget control (the total update is proportionally allocated according to local uncertainty). After N_{steps} iterations, the refined representation $\mathbf{x}_0$ is passed to the decoder to produce the final prediction.

3.5. Joint Training Objective with Output-Space Consistency

Our model comprises two complementary branches: (i) a discriminative decoder that predicts a set of moment candidates via joint classification and temporal span regression, and (ii) a generative diffusion branch that operates in the representation space. The two branches are tied together by a light consistency alignment that matches decoder outputs between “clean” and “refined” memories on positive queries, reducing the train–inference distribution gap and preventing drift.

Discriminative main loss. The decoder generates K queries and establishes a one-to-one correspondence between each query and its annotated segment via Hungarian matching. Based on this assignment and following the overall design of [31, 32], we define the discriminative main loss as

$$\mathcal{L}_{\text{main}} = \mathcal{L}_{\text{hl}} + \mathcal{L}_{\text{mr}} + \mathcal{L}_{\text{neg}}. \quad (13)$$

Here, $\mathcal{L}_{\text{mr}}$ denotes the moment retrieval loss, which combines ℓ_1 , GIoU, and classification terms to accurately regress the temporal boundaries of target segments; $\mathcal{L}_{\text{hl}}$ represents the highlight detection loss that incorporates margin ranking and rank-aware contrastive constraints to model clip-level saliency; and $\mathcal{L}_{\text{neg}}$ employs a negative-pair learning mechanism to suppress responses from irrelevant video–text pairs, thereby enhancing the model’s discriminative ability and semantic consistency.

Consistency alignment with task-oriented reconstruction. Beyond decoding from the “clean” memory, we also feed the task-oriented reconstruction $\hat{\mathbf{x}}_0$ from Eq. (10) into the same decoder to produce an auxiliary prediction. To register the generative reconstruction with the discriminative decoder in the *output space*, we impose a lightweight consistency regularizer only on the set of positively matched queries $\mathcal{Q}^+$. Let $\varphi_{\text{cls, clean}}(q)$ denote the decoder outputs (classification logits and center/scale regression) from the clean memory, and $\varphi_{\text{ref}}(q)$ the corresponding outputs when decoding $\hat{\mathbf{x}}_0$. The loss is:

$$\begin{aligned} \mathcal{L}_{\text{cons}} = \frac{1}{|\mathcal{Q}^+|} \sum_{q \in \mathcal{Q}^+} & \left(\|\varphi_{\text{cls, clean}}(q) - \varphi_{\text{cls, ref}}(q)\|_1 \right. \\ & \left. + \alpha \|\varphi_{\text{reg, clean}}(q) - \varphi_{\text{reg, ref}}(q)\|_1 \right), \end{aligned} \quad (14)$$

where α balances the relative strengths of classification and regression alignment. This term functions as an output-distribution distillation for the reconstructed branch: it encourages adherence to the VMR-oriented discriminative submanifold induced by Eq. (10) while keeping the supervision and optimization dominance of the main branch unchanged, and—by restricting to $\mathcal{Q}^+$ —avoids noise from unmatched queries.

Training Loss. The total loss of our model comprises three terms:

$$\mathcal{L} = \lambda_{\text{main}} \mathcal{L}_{\text{main}} + \lambda_{\text{diff}} \mathcal{L}_{\text{diff}} + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}}. \quad (15)$$

where λ_{main} , λ_{diff} , λ_{cons} are the weights of the main, diffusion, and consistency losses, respectively.

4. Experiments

4.1. Dataset and Evaluation Metrics

To assess the effectiveness of our approach, we conduct comprehensive experiments on three widely used benchmarks: QVHighlights [16], Charades-STA [6], and TACoS [37]. Unless otherwise specified, video motion–semantic features and textual features are extracted offline using publicly available pretrained SlowFast and CLIP models and are kept fixed throughout training and evaluation. To ensure fair comparison, we adopt the same feature configurations, training hyperparameters, and official evaluation protocols as in prior work [17, 29, 31, 32, 49], unless otherwise specified. For the VMR task, we report Recall@1 at IoU = 0.5 and 0.7, as well as mean Average Precision (mAP) averaged across multiple IoU thresholds. For highlight detection (HD), we evaluate using mAP and HIT@1.

4.2. Experimental Results

Table 1 reports joint MR/HD evaluation on QVHighlights, with results on both the test and val splits. The comparison spans traditional matching/ranking methods, the DETR family, and recent generative/denoising paradigms. ConfDiff (ours) attains consistent state-of-the-art performance across both tasks, with particularly stable advantages on high-threshold metrics (R1@0.7, mAP@0.75) and on HD’s “≥ Very Good” mAP and HIT@1. We attribute these gains to confidence-guided selective diffusion and discriminative–generative consistency alignment, which jointly improve localization robustness near ambiguous boundaries and yield better score calibration.

Table 2 compares moment retrieval across TACoS and Charades-STA. ConfDiff attains state-of-the-art (or tied) overall performance, achieving mIoU of 38.23 and 50.35, respectively, while remaining first-tier under stringent IoU thresholds (e.g., R@0.7). Notably, our approach surpasses classical matching/ranking and DETR-style baselines and further outperforms the recent LLM-based LLMPEP, indicating stronger robustness to fuzzy boundaries and improved score calibration.

4.3. Ablation study

The contribution of each major component is summarized in Table 3, where ConfDiff (full) consistently achieves the best performance across all MR metrics, particularly under strict thresholds. Removing Frame-ConfMap noticeably

Table 1. QVHighlights val/test performance. We report mean mAP computed across IoU thresholds from 0.5 to 0.95 in increments of 0.05.

Split	test							val						
Method	MR					HD		MR					HD	
	R1		mAP			≥ Very Good		R1		mAP			≥ Very Good	
	@0.5	@0.7	@0.5	@0.75	Avg	mAP	HIT@1	@0.5	@0.7	@0.5	@0.75	Avg	mAP	HIT@1
BeautyThumb [38]	-	-	-	-	-	14.36	20.88	-	-	-	-	-	-	-
DVSE [23]	-	-	-	-	-	18.75	21.79	-	-	-	-	-	-	-
MCN [1]	11.41	2.72	24.94	8.22	10.67	-	-	-	-	-	-	-	-	-
CAL [4]	25.49	11.54	23.40	7.65	9.89	-	-	-	-	-	-	-	-	-
XML [15]	41.83	30.35	44.63	31.73	32.14	34.49	55.25	-	-	-	-	-	-	-
XML+ [15]	46.69	33.46	47.89	34.67	34.90	35.38	55.06	-	-	-	-	-	-	-
Moment-DETR [16]	52.89	33.02	54.82	29.40	30.73	35.69	55.60	53.94	34.84	-	-	32.20	35.65	55.55
UMT [24]	56.23	41.18	53.38	37.01	36.12	38.18	59.99	60.26	44.26	-	-	38.59	39.85	64.19
QD-DETR [32]	62.40	44.98	62.52	39.88	39.86	38.94	62.40	62.68	46.66	62.23	41.82	41.22	39.13	63.03
UniVGT [18]	58.86	40.86	57.60	35.59	35.47	38.20	60.96	59.74	-	-	-	36.13	38.83	61.81
EaTR [12]	-	-	-	-	-	-	-	61.36	45.79	61.86	41.91	41.74	37.15	58.65
CG-DETR [31]	65.43	48.38	64.51	42.77	42.86	40.33	66.21	67.35	52.06	65.57	45.73	44.93	40.79	66.71
TR-DETR [39]	64.66	48.96	63.98	43.73	42.62	39.91	63.42	-	-	-	-	-	-	-
MS-DETR [29]	64.72	48.77	66.41	44.91	44.89	40.45	65.95	66.9	51.68	67.19	46.05	46.00	40.57	66.58
MomentDiff [17]	57.42	39.66	54.02	35.73	35.95	-	-	-	-	-	-	-	-	-
DenoiseLoc [43]	59.27	45.07	61.30	43.07	42.96	-	-	-	-	-	-	-	-	-
DiffusionVMR [49]	62.91	47.08	63.68	43.88	42.74	39.08	65.43	-	-	-	-	-	-	-
FVE [26]	63.35	47.16	62.17	42.00	41.33	-	-	-	-	-	-	-	-	-
ConfDiff (ours)	64.97	48.30	63.79	42.65	42.26	40.23	66.14	67.05	51.89	65.85	45.83	44.03	40.72	66.30

Table 2. Results on the MR datasets TACoS and Charades-STA. Video features are extracted using SlowFast and CLIP.

Method	Year	TACoS				Charades-STA			
		R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU
2D-TAN [48]	2020	40.01	27.99	12.92	27.22	58.76	46.02	27.50	41.25
VSLNet [46]	2020	35.54	23.54	13.15	24.99	60.30	42.69	24.14	41.58
Moment-DETR [16]	2021	37.97	24.67	11.97	25.49	65.83	52.07	30.59	45.54
QD-DETR [32]	2023	-	-	-	-	-	57.31	32.55	-
LLaViLo [30]	2023	-	-	-	-	-	55.72	33.43	-
UniVTG [18]	2023	51.44	34.97	17.35	33.60	70.81	58.01	35.65	50.10
CG-DETR [31]	2023	52.23	39.61	22.23	36.48	70.43	58.44	36.34	50.13
TR-DETR [39]	2024	-	-	-	-	-	57.61	33.52	-
LLMEPET [13]	2024	52.73	-	22.78	36.55	70.91	-	36.49	50.25
RGTR [40]	2025	53.04	40.31	24.32	37.44	72.04	57.93	35.16	50.30
MomentDiff [17]	2023	44.78	33.68	-	-	-	55.57	32.42	-
DenoiseLoc [43]	2023	47.36	35.89	-	-	-	-	-	-
DiffusionVMR [49]	2024	47.41	36.79	-	-	-	57.74	37.23	-
FVE [26]	2025	48.09	31.92	-	31.61	-	52.37	31.94	44.59
ConfDiff (ours)	2025	52.85	40.26	24.17	38.23	71.34	58.30	35.89	50.35

Table 3. Effects of different components on the QVHighlights validation split.

Variant	mAP			R1		$\geq$ Very Good	
	@0.5	@0.75	Avg	@0.5	@0.7	mAP	HIT@1
ConfDiff (full)	65.85	45.83	44.03	67.05	51.89	40.72	66.30
w/o Frame-ConfMap	63.79	43.62	41.72	64.62	48.43	39.63	65.21
w/o Selective Diffusion	62.87	42.78	40.67	63.67	47.17	39.51	64.55
w/o Output Consistency	60.23	43.24	43.76	60.89	46.24	38.36	63.26

weakens boundary discrimination, confirming the importance of confidence-guided cross-modal aggregation. Disabling Selective Diffusion further degrades high-threshold metrics, as uniform correction tends to over-adjust confident regions. Excluding Output Consistency causes a moderate yet consistent drop, especially on discriminative metrics, suggesting that lightweight consistency helps align refinement and decoding objectives.

Under a budget-aligned setting ($|M|$), Table 4 shows that Low-Confidence consistently surpasses Random in cal-

Table 4. Effect of the masking strategy on performance.

Masking Strategy	mAP			R1		$\geq$ Very Good	
	@0.5	@0.75	Avg	@0.5	@0.7	mAP	HIT@1
Random ($ \mathcal{M} $ matched)	64.18	44.47	43.17	66.32	50.57	39.31	63.77
All-Frames Diffusion	63.24	43.66	41.13	65.67	50.89	39.05	62.79
Oracle Border Mask	66.15	46.12	44.26	67.84	52.32	41.22	67.28
Low-Confidence	65.85	45.83	44.03	67.05	51.89	40.72	66.30

Table 5. The impact of the refinement setting on performance.

Refinement Setting	mAP			R1		$\geq$ Very Good	
	@0.5	@0.75	Avg	@0.5	@0.7	mAP	HIT@1
No Ref. ($N=0$)	62.21	41.65	40.23	64.68	47.71	38.59	61.32
Fixed: $N=3$, $u_0=0.2$	63.65	43.42	42.35	65.29	49.48	39.33	62.41
Fixed: $N=5$, $u_0=0.2$	64.18	44.68	43.14	66.43	50.24	39.61	63.35
Adapt.: $N=3$, $u_i \propto (1-\hat{c}_i)$	65.85	45.83	44.03	67.05	51.89	40.72	66.30
Adapt.: $N=5$, $u_i \propto (1-\hat{c}_i)$	65.96	45.57	44.19	67.41	51.43	40.36	66.44

ibration metrics at stringent thresholds and on HD, indicating that confidence-map-guided, uncertainty-aware selection effectively steers limited generative capacity toward genuinely ambiguous frames and improves both boundary fitting and score calibration. By contrast, All-Frames applies uniform updates, diluting the budget and inducing over-smoothing and score drift, which noticeably degrades performance at high thresholds and on HIT@1. For context, the Oracle Border Mask—activating diffusion only near GT boundaries—achieves only slightly higher curves than ours, suggesting that the main remaining gains come from further refining the confidence map rather than enlarging the mask or intensifying refinement.

As shown in Tab. 5, introducing a small number of deterministic refinement steps at inference yields consistent gains over the no-refinement baseline. The uncertainty-aware schedule, which scales each position’s update by $1 - \hat{c}_i$, further surpasses fixed-step updates on strict-threshold MR metrics (R1@0.7, mAP@0.75) and on HD calibration (mAP@($\geq$ VG), HIT@1). This substantiates the principle of refine more where we are less certain: concentrating the limited update budget on low-confidence frames improves boundary fitting while preserving score calibration. Increasing the number of steps beyond a modest budget (e.g., from $N=3$ to $N=5$) brings diminishing returns, outlining a clear efficiency–accuracy trade-off without introducing instability.

4.4. Qualitative Results

In Fig. 3, we present qualitative comparisons on QVHighlights for both highlight detection and moment retrieval. Compared with the baseline [17], ConfDiff produces temporally tighter MR segments with higher overlap to the ground truth, together with frame-level saliency curves that more faithfully track human annotations. In the first example, MomentDiff drifts beyond the true highlight, yield-

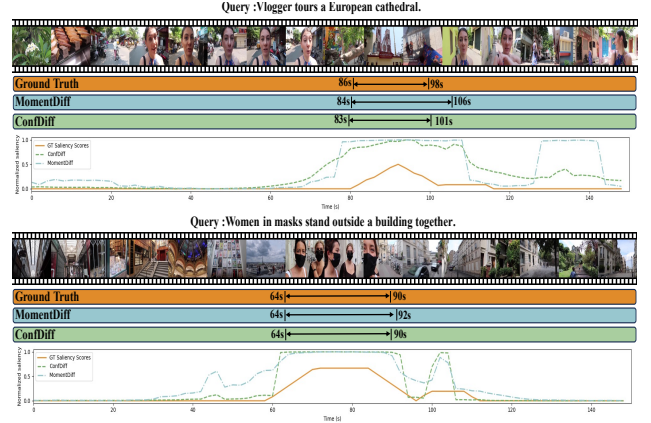


Figure 3. Qualitative comparison of predictions on QVHighlights video clips, showing sampled frames, ground-truth moments, and temporal segments predicted by the baseline and our ConfDiff, along with their normalized saliency curves.

ing an overly broad plateau and a spurious secondary peak, whereas ConfDiff concentrates probability mass on the core event window and closely follows the GT rise–plateau–decay pattern. A similar behavior is observed in the second example: ConfDiff captures both the onset and offset of the target moment while keeping non-salient regions suppressed, in contrast to the early and lingering activations of MomentDiff. These examples indicate that confidence-guided diffusion offers more reliable boundary modeling and better-calibrated highlight scores.

5. Conclusion

In this work, we introduce ConfDiff, a diffusion-based framework that reinterprets video moment retrieval from a generative perspective. By modeling temporal boundaries as stochastic representations rather than deterministic points, ConfDiff explicitly captures the uncertainty and multi-modality inherent in real-world event transitions. Built upon a confidence-guided correlation graph, our approach localizes ambiguous temporal regions and selectively applies diffusion in the representation space, enabling the model to refine more where it is less certain. In addition, a lightweight output-space consistency module bridges the gap between generative refinement and discriminative decoding, yielding stable optimization and well-calibrated confidence estimates. Extensive experiments validate the robustness and effectiveness of our framework across different levels of temporal ambiguity. We hope ConfDiff will broaden the frontier of distribution-aware temporal understanding and inspire future work toward unifying uncertainty modeling with discriminative reasoning in multi-modal video reasoning.

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